Appendix of Submission 5783
LagAttention: Modeling Temporal Asynchrony in Multivariate Time Series Forecasting

Anonymous submission

A. Methodology

A.1 Temporal-Channel Cross Attention

LagAttention is the first work that focuses on temporal-channel cross attention among variates of every time period. In this part, we further specify the attention mechanisms, including temporal-domain attention, channel-domain attention, hybird attention and temporal-channel cross attention.

Temporal-domain Attention. As shown in Fig. 1(a), the temporal-domain attention mainly explores the data variation along the time, especially the *intraperiod* and *interperiod* variations by dividing a time series into multiple periods, giving thorough connection analysis in the time domain.

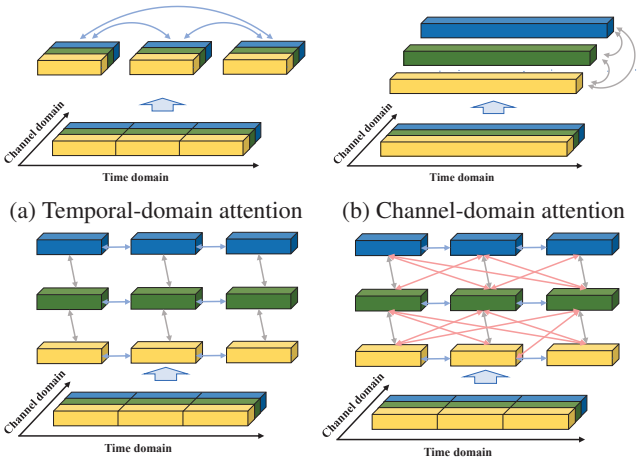
Channel domain attention. As shown in Fig. 1(b), the channel-domain attention aims to discover the correlations among variates in the channel domain. Channel-domain attention-based approaches have a basic assumption that different variates are not independent.

Hybird Attention. As shown in Fig. 1(c), the hybrid attention combines temporal attention (represented by the blue lines) and channel attentions (represented by the grey lines), capturing their respective information. The hybrid-attention-based solutions treat temporal and channel attention separately, *i.e.*, only consider the correlations among vairates in the same time patch or sub-patches in the same variate.

Temporal-Channel Cross Attention (TCCA). In comparison, as shown in Fig. 1(d), LagAttention combines the temporal and channel domains to allow any patch of data to find correlations with others in any period of any variate (the pink lines). Compared with previous attention mechanisms, Temporal-Channel Cross Attention (TCCA) can adapt to macro-asynchronies introduced by practical factors, making it more compatible to the real-world applications.

A.2 Variate Clustering

Previous appraoches often apply channel-domain attention to capture possible correlations among variates. However, variates often reflect different physical measurements, making the occasional correlations unreliable. In our design, we propose a mechanism called Grouped Synchronization (GS), which first classifies variates into multiple clusters to mitigate noise from spurious correlations while preserving genuine dependencies. To verify that variates of the same cluster indeed have higher correlations while others are not,



(a) Temporal-domain attention (b) Channel-domain attention

(c) Hybird attention (d) Temporal-Channel attention

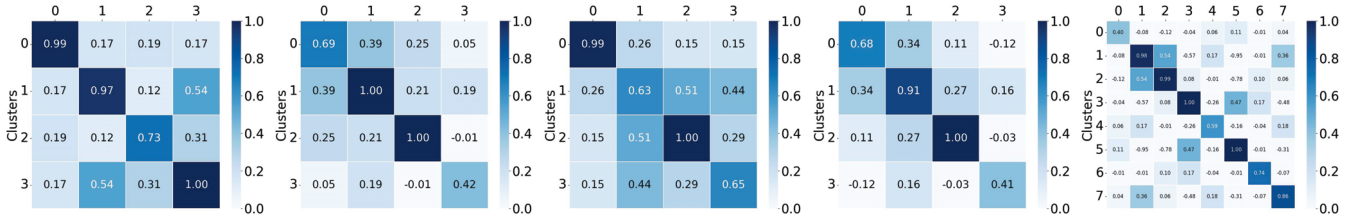
Figure 1: Attention mechanisms.

we exhibit the average cosine similarities within or among clusters in Fig. 2. We conduct experiments in five typical long-term time series datasets. The results show that the cosine similarity within a cluster is much higher than that between clusters, indicating that considering variate correlations before alignment is necessary. We also conduct ablation experiments to evaluate the effectiveness of clustering. The results exhibited in Tab. 4 show that in all the six real-world datasets, our method have better performance, indicating that our method can mitigate noise introduced by irrelevant variates.

B. Experiment and Evaluation

B.1 Experimental Setting

As shown in Tab.1, we conduct experiments on 14 real-world datasets to assess the performance of LagAttention, including **nine** long-term datasets, *i.e.*, ETT (ETTh1, ETTh2, ETTm1, ETTm2) (Zhou et al. 2021), Electricity(Trindade 2011), Weather(Huber et al. 2025), Exchange (Lai et al. 2018), Traffic (Zhao 2020), and Solar (Kumar 2020), and **five** short-term datasets, *i.e.*, ILI (Yang et al. 2020), PEMS03, PEMS04, PEMS07, PEMS08(Guo et al. 2019). These datasets cover several real-world scenarios, including energy, traffic, weather, health-care, *etc.*, making our evaluation more thorough.



(a) ETTh1 (b) ETTh2 (c) ETTm1 (d) ETTm2 (e) Weather

Figure 2: The effectiveness of variate clustering. The value in the (x, y) coordinate exhibits the average cosine similarity between the x-th cluster with the y-th cluster. The larger the similarity, the darker the color shows.

Table 1: The Statistics of datasets used in our experiments.

Dataset	Long-term dataset							Short-term dataset				
	ETTh1&2	ETTm1&2	ELC	Weather	Exchange	Traffic	Solar	ILI	PEMS03	PEMS04	PEMS07	PEMS08
Variates	7	7	321	21	8	862	137	7	358	307	883	170
Timesteps	17,420	69,680	26,304	52,696	7,588	17,544	52560	966	26208	16992	28224	17856
Granularity	1 hour	15 min	1 hour	10 min	1 day	1 hour	10min	1 week	5 min	5 min	5 min	5 min

Table 2: Short-term Time Series Forecasting (The look-back window $T = 96$ and the prediction length is $D = 24$)

Method	Metrics	Ours	iTransformer (2024)	TiDE (2023)	PatchTST (2023)	DLinear (2023)	FEDformer (2022)	FiLM (2022)	Autoformer (2021)	Informer (2021)	Stationary (2021)
ILI	MSE	2.083	-	-	**1.319**	2.398	3.228	-	3.483	5.764	-
	MAE	0.879	-	-	**0.754**	1.040	1.260	-	1.287	1.677	-
PEMS03	MAE	19.17	18.13	18.39	18.95	19.70	19.00	21.36	18.08	19.19	**17.64**
	MAPE	19.98	19.19	17.39	**17.29**	18.35	18.57	18.35	18.75	19.58	17.56
	RMSE	29.85	28.86	30.59	30.15	32.35	30.05	35.07	**27.82**	32.7	28.37
PEMS04	MAE	25.05	23.42	23.99	24.86	24.62	26.51	26.74	25.00	**22.05**	22.34
	MAPE	16.21	17.83	**14.76**	16.65	16.12	16.76	16.46	16.70	14.88	14.85
	RMSE	38.98	35.75	37.30	40.46	39.51	41.81	42.86	38.02	36.20	**35.47**
PEMS07	MAE	28.92	**22.54**	25.31	27.87	28.65	27.92	28.76	26.92	27.26	26.02
	MAPE	12.52	12.77	**11.02**	12.69	12.15	12.29	11.21	11.83	11.63	11.75
	RMSE	44.47	**33.92**	39.12	42.56	45.02	42.29	45.85	40.60	45.81	42.34
PEMS08	MAE	19.57	**18.79**	19.73	20.35	20.26	20.56	22.11	20.47	20.96	19.29
	MAPE	**11.72**	12.19	12.02	13.15	12.09	12.41	12.81	12.27	13.20	12.21
	RMSE	31.20	**28.86**	30.67	31.04	32.38	32.97	35.13	31.52	30.61	38.62

B.2 Main Results

B.2.1 Long-term Time Series Forecasting

Baselines. We choose **15 state-of-the-art models** to serve as long-term time series forecasting baselines, including SimpleTM (Chen et al. 2025), Amplifier (Fei et al. 2025), S-Mamba (Wang et al. 2025), Fredformer (Piao et al. 2024), TimesMixer (Wang et al. 2024a), TimeXer (Wang et al. 2024b), iTransformer (Liu et al. 2023), SOFTS (Han et al. 2024), PatchTST (Nie et al. 2022), TimesNet (Wu et al. 2022), Crossformer (Zhang and Yan 2023)), DLinear (Zeng et al. 2023), FEDformer (Zhou et al. 2022b), Autoformer (Wu et al. 2021) and Informer (Zhou et al. 2021). To ensure fair comparisons, all models follow the same input length ($T = 96$) and prediction length ($L \in \{96, 192, 336, 720\}$).

Performance. Table 3 shows the prediction results. The best ones are in red and the second are in blue. The lower MSE/MAE indicates the more accurate prediction result. Obviously, LagAttention stands out with the top two places in **47 cases** out of the overall **72 cases** and achieves the best performance in five datasets over eight long-term time series forecasting tasks. The most recent work, *i.e.*, SimpleTM(Chen et al. 2025), also has a good performance, which win the top two places in **44 cases** and achieves the best performance in two datasets.

B.2.2 Short-term Time Series Forecasting

Baselines. We choose **9** state-of-the-art models to serve as baselines for short-term time series forecasting baselines, including iTransformer (Liu et al. 2023), TiDE (Das et al. 2023), PatchTST (Nie et al. 2022), DLinear (Zeng et al. 2023), FEDformer (Zhou et al. 2022b), FiLM (Zhou et al. 2022a), Autoformer (Wu et al. 2021), Informer (Zhou et al. 2021) and Stationary (Liu et al. 2022). We select two common metrics for the ILI dataset, *i.e.*, Mean Absolute Error (MAE) and Mean Squared Error (MSE). We use three matrics for the four PESMS datasets, *i.e.*, Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE), where lower values indicate better prediction.

Performance. Tab. 2 exhibit the performance of LagAttention in five datasets. We find that LagAttention is not

Table 3: Multivariate forecasting results with prediction lengths $D \in \{96, 192, 336, 720\}$ and fixed look-back length $T = 96$.

Models	D	Metrics	**LagAttention (Ours)**	SimpleTM (2025)	Amplifier (2025)	S-Mamba (2025)	Fredformer (2024)	TimeMixer (2024)	TimeXer (2024)	iTransformer (2024)	SOFTS (2024)	PatchTST (2023)	TimesNet (2023)	Crossformer (2023)	DLinear (2023)	FEDformer (2022)	Autoformer (2021)	Informer (2021)
ETTh1	96	MSE	**0.361**	0.368	0.371	0.386	0.373	0.375	0.377	0.386	0.381	0.460	0.384	0.423	0.386	0.395	0.449	0.865
		MAE	**0.389**	0.393	0.392	0.405	0.392	0.400	0.397	0.405	0.399	0.447	0.402	0.448	0.400	0.424	0.459	0.713
	192	MSE	**0.404**	0.422	0.426	0.443	0.433	0.429	0.425	0.441	0.435	0.512	0.436	0.471	0.437	0.469	0.500	1.008
		MAE	**0.412**	0.424	0.422	0.437	0.420	0.421	0.426	0.436	0.431	0.477	0.429	0.474	0.432	0.470	0.482	0.792
	336	MSE	**0.426**	0.439	0.448	0.489	0.470	0.484	0.457	0.487	0.480	0.546	0.491	0.570	0.481	0.530	0.521	1.107
		MAE	**0.424**	0.434	0.434	0.468	0.437	0.458	0.441	0.458	0.452	0.496	0.469	0.546	0.459	0.499	0.496	0.809
	720	MSE	**0.452**	0.460	0.476	0.502	0.467	0.498	0.464	0.503	0.499	0.544	0.521	0.653	0.519	0.598	0.514	1.181
		MAE	**0.454**	0.461	0.464	0.489	0.456	0.482	0.463	0.491	0.488	0.517	0.500	0.621	0.516	0.544	0.512	0.865
	Avg	MSE	**0.411**	0.422	0.430	0.455	0.435	0.447	0.431	0.454	0.449	0.516	0.458	0.529	0.455	0.498	0.496	1.040
		MAE	**0.420**	0.428	0.428	0.450	0.426	0.440	0.432	0.447	0.442	0.484	0.450	0.522	0.451	0.484	0.487	0.795
ETTh2	96	MSE	**0.268**	0.281	0.279	0.296	0.293	0.289	0.289	0.297	0.297	0.308	0.340	0.745	0.295	0.358	0.346	3.755
		MAE	**0.324**	0.337	0.337	0.348	0.342	0.341	0.340	0.349	0.347	0.355	0.374	0.584	0.352	0.397	0.388	1.525
	192	MSE	**0.335**	0.351	0.359	0.376	0.371	0.372	0.370	0.380	0.373	0.393	0.402	0.877	0.452	0.429	0.456	5.602
		MAE	**0.372**	0.383	0.389	0.396	0.389	0.392	0.391	0.400	0.394	0.405	0.414	0.656	0.462	0.439	0.452	1.931
	336	MSE	**0.338**	0.363	0.377	0.424	0.382	0.386	0.422	0.428	0.410	0.427	0.452	1.043	0.504	0.496	0.482	4.721
		MAE	**0.383**	0.400	0.406	0.431	0.409	0.414	0.434	0.432	0.426	0.436	0.452	0.731	0.490	0.487	0.486	1.835
	720	MSE	**0.398**	0.414	0.420	0.426	0.415	0.412	0.429	0.427	0.411	0.436	0.462	1.104	0.577	0.463	0.515	3.647
		MAE	**0.426**	0.436	0.432	0.444	0.434	0.434	0.445	0.445	0.433	0.450	0.468	0.763	0.538	0.474	0.511	1.625
	Avg	MSE	**0.335**	0.352	0.359	0.381	0.365	0.364	0.378	0.383	0.373	0.391	0.414	0.942	0.457	0.437	0.450	4.431
		MAE	**0.376**	0.389	0.391	0.405	0.393	0.395	0.403	0.407	0.400	0.411	0.427	0.684	0.460	0.449	0.459	1.729
ETTm1	96	MSE	0.310	0.314	0.316	0.333	0.326	0.320	**0.309**	0.334	0.325	0.352	0.338	0.404	0.345	0.379	0.505	0.672
		MAE	**0.348**	0.352	0.355	0.368	0.361	0.357	0.352	0.368	0.361	0.374	0.375	0.426	0.372	0.419	0.475	0.571
	192	MSE	0.356	0.360	0.361	0.376	0.363	0.361	**0.355**	0.377	0.375	0.390	0.374	0.450	0.380	0.426	0.553	0.795
		MAE	**0.375**	0.379	0.381	0.390	0.380	0.381	0.378	0.391	0.389	0.393	0.387	0.451	0.389	0.441	0.496	0.669
	336	MSE	**0.387**	0.392	0.393	0.408	0.395	0.390	**0.387**	0.426	0.405	0.421	0.410	0.532	0.413	0.445	0.621	1.212
		MAE	**0.397**	0.403	0.402	0.413	0.403	0.404	0.399	0.420	0.412	0.414	0.411	0.515	0.413	0.459	0.537	0.871
	720	MSE	0.450	0.451	0.455	0.475	0.453	0.454	**0.448**	0.491	0.466	0.462	0.478	0.666	0.474	0.543	0.671	1.166
		MAE	**0.434**	0.440	0.440	0.448	0.438	0.441	0.435	0.459	0.447	0.449	0.450	0.589	0.453	0.490	0.561	0.823
	Avg	MSE	0.376	0.379	0.381	0.398	0.384	0.381	**0.375**	0.407	0.393	0.406	0.400	0.513	0.403	0.448	0.588	0.961
		MAE	**0.389**	0.394	0.394	0.405	0.395	0.395	0.391	0.410	0.403	0.407	0.406	0.495	0.406	0.452	0.517	0.734
ETTm2	96	MSE	**0.171**	0.174	0.176	0.179	0.177	0.175	**0.171**	0.180	0.180	0.183	0.187	0.287	0.183	0.204	0.255	0.365
		MAE	**0.253**	0.256	0.258	0.263	0.259	0.258	0.255	0.264	0.261	0.270	0.267	0.366	0.273	0.288	0.339	0.453
	192	MSE	0.239	0.242	0.239	0.250	0.241	**0.237**	0.238	0.250	0.246	0.255	0.249	0.414	0.260	0.316	0.281	0.533
		MAE	**0.297**	0.302	0.300	0.309	0.300	0.299	0.300	0.309	0.306	0.314	0.309	0.492	0.325	0.363	0.340	0.563
	336	MSE	0.297	**0.296**	0.297	0.312	0.302	0.298	0.301	0.311	0.319	0.309	0.321	0.597	0.336	0.359	0.339	1.363
		MAE	**0.336**	0.337	0.338	0.349	0.340	0.340	0.340	0.348	0.352	0.347	0.351	0.542	0.367	0.387	0.372	0.887
	720	MSE	0.394	0.393	0.393	0.411	0.397	**0.391**	0.401	0.412	0.405	0.412	0.408	1.730	0.415	0.433	0.422	3.379
		MAE	**0.390**	0.394	0.396	0.406	0.396	0.396	0.397	0.407	0.401	0.404	0.403	1.042	0.423	0.432	0.419	1.338
	Avg	MSE	**0.275**	0.276	0.276	0.288	0.279	**0.275**	0.278	0.288	0.287	0.290	0.291	0.757	0.298	0.328	0.324	1.410
		MAE	**0.319**	0.323	0.323	0.332	0.324	0.323	0.323	0.332	0.330	0.334	0.333	0.610	0.347	0.367	0.367	0.810
Weather	96	MSE	0.162	0.161	**0.156**	0.165	0.163	0.163	0.168	0.174	0.166	0.186	0.172	0.195	0.196	0.227	0.266	0.300
		MAE	0.205	0.205	**0.204**	0.210	0.207	0.209	0.209	0.214	0.208	0.227	0.220	0.271	0.255	0.304	0.336	0.384
	192	MSE	**0.207**	0.209	0.209	0.214	0.211	0.208	0.220	0.221	0.217	0.234	0.219	0.209	0.237	0.295	0.307	0.598
		MAE	**0.247**	0.248	0.249	0.252	0.251	0.250	0.254	0.254	0.253	0.265	0.261	0.277	0.296	0.363	0.367	0.544
	336	MSE	0.261	0.265	0.264	0.274	0.267	**0.251**	0.276	0.278	0.282	0.284	0.280	0.273	0.283	0.381	0.359	0.578
		MAE	**0.286**	0.291	0.290	0.297	0.292	0.287	0.296	0.296	0.300	0.301	0.306	0.332	0.335	0.416	0.395	0.523
	720	MSE	0.342	0.344	0.343	0.350	0.343	**0.339**	0.353	0.358	0.356	0.356	0.365	0.379	0.345	0.424	0.419	1.059
		MAE	**0.339**	0.343	0.342	0.345	0.341	0.341	0.347	0.349	0.351	0.349	0.359	0.401	0.381	0.434	0.428	0.741
	Avg	MSE	0.243	0.244	0.243	0.251	0.246	**0.240**	0.254	0.258	0.255	0.265	0.259	0.264	0.265	0.331	0.337	0.634
		MAE	**0.269**	0.272	0.271	0.276	0.272	0.271	0.276	0.279	0.278	0.285	0.287	0.320	0.316	0.379	0.381	0.548
Solar	96	MSE	0.239	**0.170**	-	0.205	-	0.189	0.220	0.203	0.200	0.529	0.373	0.232	0.290	0.286	0.456	0.287
		MAE	0.272	**0.225**	-	0.244	-	0.259	0.278	0.237	0.234	0.241	0.358	0.302	0.378	0.341	0.446	0.323
	192	MSE	0.271	**0.187**	-	0.237	-	0.222	0.251	0.233	0.231	0.288	0.397	0.371	0.320	0.291	0.588	0.297
		MAE	0.302	**0.243**	-	0.270	-	0.283	0.297	0.261	0.260	0.332	0.376	0.410	0.398	0.337	0.561	0.341
	336	MSE	0.290	**0.193**	-	0.258	-	0.231	0.275	0.248	0.246	0.301	0.420	0.495	0.353	0.354	0.595	0.367
		MAE	0.321	**0.248**	-	0.288	-	0.292	0.313	0.273	0.274	0.339	0.380	0.515	0.415	0.416	0.588	0.429
	720	MSE	0.284	**0.200**	-	0.260	-	0.223	0.281	0.249	0.249	0.295	0.420	0.526	0.357	0.380	0.733	0.374
		MAE	0.313	**0.255**	-	0.288	-	0.285	0.317	0.275	0.277	0.336	0.381	0.542	0.413	0.437	0.633	0.431
	Avg	MSE	0.271	**0.188**	-	0.240	-	0.216	0.256	0.233	0.231	0.287	0.403	0.406	0.330	0.328	0.586	0.331
		MAE	0.302	**0.244**	-	0.273	-	0.280	0.301	0.262	0.261	0.333	0.374	0.442	0.401	0.383	0.557	0.381
Electricity	96	MSE	0.170	**0.137**	0.147	0.139	0.147	0.153	0.151	0.148	0.147	0.190	0.168	0.219	0.194	0.183	0.201	0.274
		MAE	0.260	**0.230**	0.243	0.235	0.241	0.247	0.247	0.240	0.237	0.296	0.272	0.314	0.276	0.297	0.317	0.368
	192	MSE	0.182	**0.152**	0.157	0.159	0.165	0.166	0.165	0.162	0.162	0.199	0.184	0.231	0.193	0.195	0.222	0.296
		MAE	0.277	**0.245**	0.251	0.255	0.258	0.256	0.261	0.253	0.251	0.304	0.289	0.322	0.280	0.308	0.334	0.386
	336	MSE	0.199	**0.171**	0.175	0.176	0.177	0.185	0.183	0.178	0.179	0.217	0.198	0.246	0.206	0.212	0.231	0.300
		MAE	0.286	**0.264**	0.269	0.272	0.273	0.277	0.280	0.269	0.269	0.319	0.300	0.337	0.296	0.313	0.338	0.394
	720	MSE	0.233	**0.200**	0.206	0.204	0.213	0.225	0.220	0.225	0.220	0.258	0.220	0.280	0.242	0.231	0.254	0.373
		MAE	0.313	**0.290**	0.296	0.298	0.304	0.310	0.309	0.317	0.305	0.352	0.320	0.363	0.329	0.343	0.361	0.439
	Avg	MSE	0.196	**0.165**	0.171	0.170	0.175	0.182	0.180	0.178	0.177	0.216	0.192	0.244	0.208	0.205	0.227	0.311
		MAE	0.284	**0.257**	0.265	0.265	0.269	0.272	0.274	0.270	0.265	0.318	0.295	0.334	0.295	0.315	0.337	0.397
Exchange	96	MSE	0.086	-	0.083	0.086	-	-	-	0.086	-	0.088	0.107	0.256	**0.078**	0.148	0.197	0.847
		MAE	0.205	-	0.202	0.207	-	-	-	0.206	-	0.205	0.234	0.367	**0.197**	0.278	0.323	0.752
Num. of Top 2 Places			**47**	44	21	5	5	19	14	3	6	0	0	0	2	0	0	0

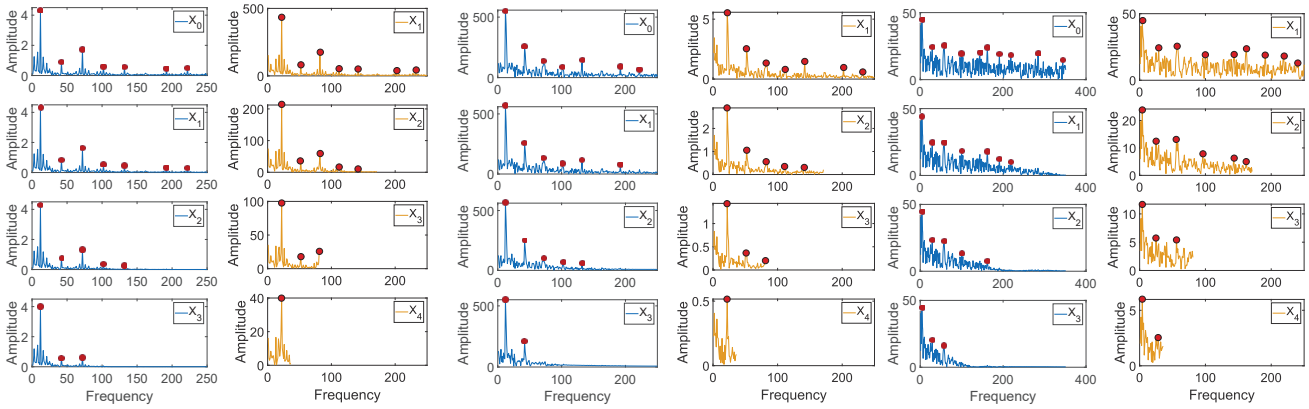
good at handling short-term time series forecasting tasks. That may because compared with long-term time series forecasting tasks, the short-term ones are more sensitive to instantaneous changes. However, our method is better at processing data with good periodicity.

B.3 Ablation Analysis

B.3.1 Module Ablation Study. To verify the effectiveness of each component of LagAttention, we provide a detailed ablation study on multiple possible designs in both the Multi-granularity Feature Refining (MFR) and Grouped Cross Attention (GCA) blocks. As depicted in Tab.4, we test six cases: 1) Removing the Iterative Filter (IF) in the MFR block; 2) Changing the IF to downsampling in the MFR block; 3) Removing the decomposation in the GCA block; 4) Removing the variate clustering in the GCA block; 5) Removing the variate alignment in the GCA block; 6) Removing the Temporal-Channel Cross Attention in the GCA block. We observe that LagAttention improves performance in all datasets, indicating the effectiveness of our designs.

Table 4: Ablation Study of LagAttention (The look-back and forecasting windows are 96).

Block	Case	ETTh1 MSE	ETTh1 MAE	ETTh2 MSE	ETTh2 MAE	ETTm1 MSE	ETTm1 MAE	ETTm2 MSE	ETTm2 MAE	Weather MSE	Weather MAE	ELC MSE	ELC MAE
Raw Model		**0.361**	**0.389**	**0.268**	**0.324**	**0.310**	**0.348**	**0.171**	**0.253**	**0.162**	**0.205**	**0.170**	**0.260**
MFR	w/o Iterative Filter	0.364	0.388	0.273	0.326	0.352	0.373	0.180	0.258	0.165	0.209	0.193	0.276
	Downsampling	0.366	0.387	0.269	0.329	0.342	0.368	0.179	0.257	0.194	0.234	0.201	0.277
GCA	w/o Decomposation	0.380	0.389	0.277	0.329	0.339	0.366	0.178	0.256	0.164	0.208	0.190	0.266
	w/o Clustering	0.366	0.388	0.273	0.329	0.315	0.353	0.181	0.258	0.166	0.210	0.176	0.268
	w/o Alignment	0.371	0.392	0.284	0.333	0.321	0.353	0.178	0.256	0.164	0.208	0.175	0.266
	w/o TCCA	0.371	0.389	0.273	0.328	0.324	0.355	0.176	0.257	0.163	0.206	0.178	0.266



(a) ELC: IF (b) ELC: DS (c) ETTh1: IF (d) ETTh1: DS (e) Weather: IF (f) Weather: DS

Figure 3: The comparison of FFT curves after the i-th Iterative Filter (IF) and Downsampling (DS) in three mainstream datasets. The red points refer to some dominant frequency components in the frequency domains.

B.3.2 Analysis of Iterative Filter. In our design of the MFR block, we apply an iterative filter instead of downsampling to refine features in varying granularities. To visualize its effectiveness, we conduct a short-time Fourier Transform (FFT) on the series X_l and show the results in Fig.3. The blue lines are the FFT results after our iterative filter (IF), and the orange lines are that after downsampling (DS). The red points refer to some dominant frequency components. According to the results, we have two observations: First, IF has better performance than DS in gradually eliminating irregular noises while maintaining dominant features. Second, IF is effective in varying datasets despite their timesteps, granularities, or other characteristics.

B.3.3 Analysis of the Clustering Mechanism. We employ a popular clustering method, *i.e.*, K-Means, to classify variates. We also compare it with other clustering approaches like DBSCAN and Hierarchical clustering (HC). As shown in Tab. 5-(a), hierarchical clustering performs better than DBSCAN and exhibits good forecasting accuracy.

Besides the clustering method, we discuss the anchor variate selection mechanism in Tab. 5-(b). In our design, we select the variate that is closest to the cluster center as the anchor variate. If we choose a random variate in the group as the anchor variate, its performance will drop significantly, indicating that an appropriate anchor variate is necessary.

B.3.4 Analysis of the Alignment Mechanism. We apply cosine similarity to align two mismatched variates in the GCA block. In this experiment, we also compare different alignment mechanisms in Tab. 6. Specifically, we use two random alignment methods, which randomly determine the time delay parameter δ using uniform and exponential distributions. We find that our cosine-similarity-based alignment mechanism outperforms other random-based approaches, indicating that tackling the micro-asynchronies between variates is crucial to the performance.

We also discuss the performance under different digest center t_0, digest length t, and patch length $\frac{T'}{P}$. In our experiments, the default setting is $t_0 = 24$, $t = 12$, and $\frac{T'}{P} = 48$. Fig. 4 exhibits the error rates (MSE and MAE) of four ETT datasets. The results indicate that the choice of digest center has little effect on the performance, while the digest length t and patch length $\frac{T'}{P}$ are crucial parameters in our model.

B.4 Performance Analysis

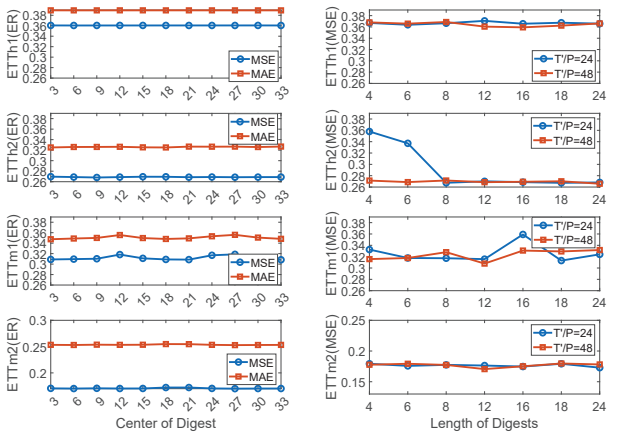
B.4.1 Scale and Efficiency The Temporal-Channel Cross Attention (TCCA) may increase the scale of the neural network. As shown in the Tab. 7, compared with other transformer-based methods, LagAttention reduces $12.36\%/12.45\%$ MSE/MAE with only 3.32% parameter costs in average.

Table 5: Analysis of the Clustering Mechanism (The look-back and forecasting windows are 96).

Case	Method	ETTh1		ETTh2		ETTm1		ETTm2	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Raw Model		**0.361**	**0.389**	**0.268**	**0.324**	**0.310**	**0.348**	**0.171**	**0.253**
(a)	DBSCAN	0.365	0.394	0.282	0.334	0.313	0.352	0.171	0.254
	HC	**0.361**	**0.389**	0.269	0.326	**0.310**	**0.348**	**0.171**	**0.253**
(b)	Random	0.363	0.389	0.274	0.330	0.313	0.350	0.171	0.254

Table 6: Analysis of the Alignment Mechanism (The look-back and forecasting windows are 96).

Method	ETTh1		ETTh2		ETTm1		ETTm2	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Raw Model	**0.361**	**0.389**	**0.268**	**0.324**	**0.310**	**0.348**	**0.171**	**0.253**
Uniform distribution	0.365	0.394	0.280	0.333	0.315	0.351	0.176	0.258
Exponential distribution	0.362	0.390	0.270	0.327	0.314	**0.347**	0.173	0.253

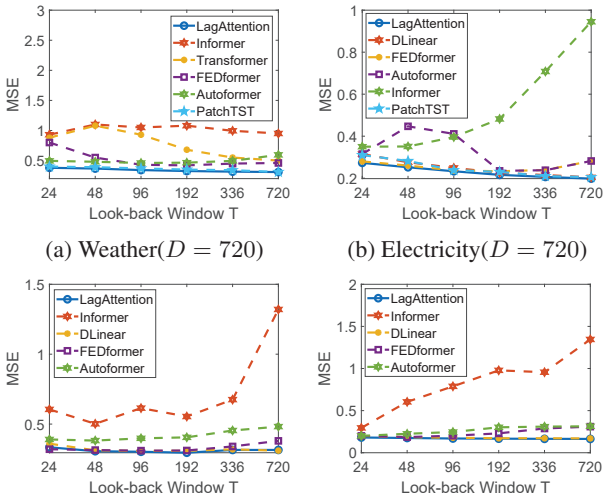


(a) Performance under different digest center t_0

(b) Performance under different digest length t

Figure 4: The performance under different digest center t_0, digest length t, and patch length $\frac{T'}{P}$.

B.4.2 Performance under Various Look-back Windows. A longer look-back window L increases the receptive field, which may improves the forecasting performance. As shown in Fig. 5, we conduct experiments with various look-back window of $L \in \{24, 48, 96, 192, 336, 720\}$ to forecast either short-term time series ($D = 24$) and long-term time series ($D = 720$). LagAttention outperforms others and have good performance in both cases.



(a) Weather($D = 720$)

(b) Electricity($D = 720$)

(c) ETTh1($D = 24$)

(d) ETTh2($D = 24$)

Figure 5: Performance under Various Look-back Windows

References

Chen, H.; Luong, V.; Mukherjee, L.; and Singh, V. 2025. SimpleTM: A Simple Baseline for Multivariate Time Series Forecasting. In *The Thirteenth International Conference on Learning Representations*.

Das, A.; Kong, W.; Leach, A.; Mathur, S.; Sen, R.; and Yu, R. 2023. Long-term forecasting with tide: Time-series dense encoder. *arXiv preprint arXiv:2304.08424*.

Fei, J.; Yi, K.; Fan, W.; Zhang, Q.; and Niu, Z. 2025. Amplifier: Bringing Attention to Neglected Low-Energy Components in Time Series Forecasting. *arXiv preprint arXiv:2501.17216*.

Guo, S.; Lin, Y.; Feng, N.; Song, C.; and Wan, H. 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, 922–929.

Han, L.; Chen, X.-Y.; Ye, H.-J.; and Zhan, D.-C. 2024. SOFTS: Efficient Multivariate Time Series Forecasting with Series-Core Fusion. *arXiv preprint arXiv:2404.14197*.

Huber, F.; Van Kuppevelt, D.; Steinbach, P.; Sauze, C.; Liu, Y.; and Weel, B. 2025. Weather Dataset. https://www.bgc-jena.mpg.de/wetter/.

Kumar, C. 2020. Solar Dataset. https://www.kaggle.com/datasets/chaitanyakumar12/time-series-forecasting-of-solar-energy.

Lai, G.; Chang, W.-C.; Yang, Y.; and Liu, H. 2018. Modeling long-and short-term temporal patterns with deep neural net-

Table 7: Comparison of method efficiency on the Electricity dataset with a look-back window size of 96 and forecasting horizon of 720 steps. (P and t are the number and length of digests within a variate. N represents the number of variate and T is the length of historical window.)

Method	Parameter	Memory	Time	Memory	MSE	MAE
LagAttention	426.11K	15140MiB	$O((Pt)^{N/K})$	$O((Pt)^{N/K})$	0.233	**0.313**
LagAttention (*w/o TCCA*)	358.14K	2490MiB	$O((Pt)^2)$	$O((Pt)^2)$	0.234	0.315
Informer	14.39M	3869MiB	$O(TlogT)$	$O(TlogT)$	0.373	0.429
Autoformer	14.91M	7607MiB	$O(TlogT)$	$O(TlogT)$	0.254	0.361
FEDformer	20.68M	4143MiB	$O(T)$	$O(T)$	**0.231**	0.343
DLinear	139.7K	687MiB	$O(T)$	$O(T)$	0.242	0.329

works. In *The 41st international ACM SIGIR conference on research & development in information retrieval*, 95–104.

Liu, Y.; Hu, T.; Zhang, H.; Wu, H.; Wang, S.; Ma, L.; and Long, M. 2023. itransformer: Inverted transformers are effective for time series forecasting. *International Conference on Learning Representations (ICLR)*.

Liu, Y.; Wu, H.; Wang, J.; and Long, M. 2022. Non-stationary transformers: Exploring the stationarity in time series forecasting. *Advances in Neural Information Processing Systems*, 35: 9881–9893.

Nie, Y.; Nguyen, N. H.; Sinthong, P.; and Kalagnanam, J. 2022. A time series is worth 64 words: Long-term forecasting with transformers. *International Conference on Learning Representations (ICLR)*.

Piao, X.; Chen, Z.; Murayama, T.; Matsubara, Y.; and Sakurai, Y. 2024. Fredformer: Frequency debiased transformer for time series forecasting. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining*, 2400–2410.

Trindade, A. 2011. ELectricity Dataset. https://archive.ics.uci.edu/ml/datasets/ElectricityLoadDiagrams20112014.

Wang, S.; Wu, H.; Shi, X.; Hu, T.; Luo, H.; Ma, L.; Zhang, J. Y.; and Zhou, J. 2024a. Timemixer: Decomposable multi-scale mixing for time series forecasting. *International Conference on Learning Representations (ICLR)*.

Wang, Y.; Wu, H.; Dong, J.; Qin, G.; Zhang, H.; Liu, Y.; Qiu, Y.; Wang, J.; and Long, M. 2024b. Timexer: Empowering transformers for time series forecasting with exogenous variables. *Advances in neural information processing systems*.

Wang, Z.; Kong, F.; Feng, S.; Wang, M.; Yang, X.; Zhao, H.; Wang, D.; and Zhang, Y. 2025. Is mamba effective for time series forecasting? *Neurocomputing*, 619: 129178.

Wu, H.; Hu, T.; Liu, Y.; Zhou, H.; Wang, J.; and Long, M. 2022. Timesnet: Temporal 2d-variation modeling for general time series analysis. *International Conference on Learning Representations (ICLR)*.

Wu, H.; Xu, J.; Wang, J.; and Long, M. 2021. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *Advances in neural information processing systems*, 34: 22419–22430.

Yang, C.-T.; Chen, Y.-A.; Chan, Y.-W.; Lee, C.-L.; Tsan, Y.-T.; Chan, W.-C.; and Liu, p.-y. 2020. Influenza-like illness prediction using a long short-term memory deep learning model with multiple open data sources. *The Journal of Supercomputing*, 76.

Zeng, A.; Chen, M.; Zhang, L.; and Xu, Q. 2023. Are transformers effective for time series forecasting? In *Proceedings of the AAAI conference on artificial intelligence*, 9, 11121–11128.

Zhang, Y.; and Yan, J. 2023. Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In *The eleventh international conference on learning representations*.

Zhao, L. 2020. Traffic Dataset. https://paperswithcode.com/dataset/traffic.

Zhou, H.; Zhang, S.; Peng, J.; Zhang, S.; Li, J.; Xiong, H.; and Zhang, W. 2021. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, 11106–11115.

Zhou, T.; Ma, Z.; Wen, Q.; Sun, L.; Yao, T.; Yin, W.; Jin, R.; et al. 2022a. Film: Frequency improved legendre memory model for long-term time series forecasting. *Advances in neural information processing systems*, 35: 12677–12690.

Zhou, T.; Ma, Z.; Wen, Q.; Wang, X.; Sun, L.; and Jin, R. 2022b. Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *International conference on machine learning*, 27268–27286. PMLR.